Supporting information – Adaptive management of ecological systems under partial observability

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Appendix A. Partially observable Markov decision process (POMDP)

POMDPs are defined as a tuple $(X, C, Z, T, O, U, \gamma, b_0)$, in which X represents the state of the system, C is available actions to the decision-maker (i.e. the agent), Z is the noisy measurements of the state, T represents the transition probability function (i.e. dynamics of the system), O is the emission probability that defines the accuracy of measurements and relates the noisy measurements to the state of the system, γ is the discount factor, relating the future economic returns to their net present value, U is the utility function, defining the economic return of managing the system for each state and action, and b_0 is the initial belief state, i.e. probability distribution over the initial state. The belief state at each time t is the posterior probability of the state space, given history of observations and taken actions, formally described as $b_t = P(x_t \mid \bar{c}_{t-1}, \bar{z}_t)$, where $\bar{z}_t = \{z_1, ..., z_t\}$ and $\bar{c}_{t-1} = \{c_1, ..., c_{t-1}\}$ indicate the history of observations and taken actions up to time t. Thus, the belief state represents the sufficient statistics in POMDP framework, meaning that given the current belief, decision making about the future is independent of past (i.e. Markov property holds on the belief space). Belief at time $t + 1$ can be updated at each time given the new observation z_{t+1}, using Bayes rule:

$$b_{t+1}(x') = \frac{O(x', c_t, z_{t+1}) \sum_{x \in X} T(x, c_t, x') b_t(x)}{\sum_{x'' \in X} O(x'', c_t, z_{t+1}) \sum_{x \in X} T(x, c_t, x'') b_t(x)} \tag{1}$$

where $T(x, c_t, x')$ means that the state at time t was x, the agent took action c_t and ended up in state x' at time $t + 1$. Similarly, $O(x', c_t, z_{t+1})$ means that the agent took action c_t at time t and ended up in state x' at time $t + 1$ and received observation z_{t+1} instead of fully observing the state.

The goal in POMDP is to find an optimal policy that maximizes the long-term economic return of managing the system. This economic return is well-known as value function, V. Hence the optimization goal is to find the maximum value of the current belief state and the corresponding policy. This can be found by solving the recursive Bellman's equation (Bellman 1957):

$$V^*(b) = \max_{c \in C} \left[\sum_{x \in X} b(x) U(x,c) + \gamma \sum_{z \in Z} P(z \mid b, c) V^*(b') \right] \tag{2}$$

where $U(x,c)$ is the economic return corresponding to the system being in state x and the agent taking action c, b' is the updated belief b_{t+1}, according to Eq.(1), with $c = c_t$, $z = z_{t+1}$, $b = b_t$, and conditional probability $P(z \mid b, c)$ is computed as

$$P(z \mid b, c) = \sum_{x' \in X} O(x', c, z) \sum_{x \in X} T(x, c, x') b(x) \tag{3}$$

Exact solutions in the belief state are theoretically possible through dynamic programming, but in practice are computationally intractable with any more than a handful of states (the curse of dimensionality) (Bertsekas 1996). Recently, new generation of POMDP solvers called point-based value iteration methods (Pineau et al. 2003; Kurniawati et al. 2008) were proposed to efficiently approximate the value function for only those belief states that are reachable given the initial condition of the system, i.e. b_0. Shani et al. (2013) provides detailed survey of these methods and their implementation. In this paper, we use SARSOP (Kurniawati et al. 2008), which is the state-of-the-art offline POMDP solver for approximating the optimal value function.

Appendix B. Optimizing decisions under model and state uncertainties

The main objective function of the decision optimization step of adaptive management, aims at maximizing the long-term expected economic return (similar to POMDP), and is formalized as follow:

$$c^*_\tau = \text{argmax}_{c_t \in C} \mathbb{E}_\Theta \left[\sum_{t=\tau}^{\infty} \gamma^t \sum_{x_t \in X} U\left(p\left(x_t\right), c_t\right) \right] \tag{4}$$

where, c^*_τ denotes the optimal harvest policy (i.e. catch) at time τ, $\mathbb{E}_y$ is the statistical expectation according to the knowledge of variable y, Θ represents the posterior of uncertain population dynamics model, given the historical measurements and the implemented policies (we show in Appendix C how this can be calculated), γ represents the discount factor, U is the utility function quantifying the economic return of implemented policy, $p(x_t)$ is the belief state (i.e. b_t): probability distribution over the population biomass conditioned on historical measurements,$\bar{z}_t = \{z_1, ..., z_t\}$, and implemented harvest policies, $\bar{c}_{t-1} = \{c_1, ..., c_{t-1}\}$, $p\left(x_t \mid \Theta, \bar{c}_{t-1}, \bar{z}_t\right)$.

As mentioned before, finding the exact solution to the objective function of Eq. (4) is intractable except for small problems. Here we develop a heuristic to overcome the computational complexity of decision making under model uncertainty and allow incorporation of both state and model uncertainties feasibly. The method is known in the family of active learning with limited reinforcement using the Bayes risk (Doshi-Velez et al. 2012; Memarzadeh et al. 2014). The heuristic we use to approximate the exact solution is that the agent finds optimal action for the current posterior distribution over parameters modeled by $P(\Theta \mid \bar{c}_{t-1}, \bar{z}_t)$, neglecting the updating attributable to future observations. To formalize this, we define $Q_\Theta(c, b)$ as the quality (or Q-value) of a belief-action pair for a POMDP, that is, the value of starting from belief b, taking action c, and following the optimal policy afterward for a model defined by Θ.

Under this construction we can define the optimal value function as follow:

$$V^*(b) = \max_{c \in C} Q^*_\Theta(c, b) \tag{5}$$

where,

$$Q^*_\Theta(c, b) = \int_\Theta \Pi(\theta) \sum_{x \in X} b(x) U(x, c) + \int_\Theta \Pi(\theta) \gamma \sum_{z \in Z} P(z \mid b, c) V^*(b') \tag{6}$$

where $\Pi_\Theta(\theta)$ is the posterior distribution of the candidate model θ. In general, this integral over the model space is intractable. Hence, we use an approximation for finding a near-optimal action that is scalable to complex problems.

Let the model-specific loss $L_\theta(c, c^*; b)$ of taking action c in model θ instead of optimal action c^* be $Q^*_\theta(c, b) - Q^*_\theta(c^*, b)$. The expected loss $E_\Theta[L]$ is the Bayes risk (Doshi-Velez et al. 2012):

$$BR\,(c, b) = \int_\Theta \left(Q^*_\theta(c, b) - Q^*_\theta(c^*, b)\right) \Pi(\theta) \tag{7}$$

where Θ is the space of models, and c^* is the optimal action in belief b_θ according to model's Q-function Q^*_θ. The first term in the integral, $(Q^*_\theta(c, b) - Q^*_\theta(c^*, b))$ is the loss incurred by taking action c instead of optimal action c^* with respect to model θ. Different models may have different optimal actions and the Bayes risk is the expected loss after averaging over models according to their posterior probabilities.

Under this definition, we can find the action with the least risk at the current time as follow,

$$c^* \cong \operatorname*{argmax}_{c \in C} \mathbb{E}_\Theta[Q_\Theta(c, b)] \tag{8}$$

where $\mathbb{E}_y$ indicates the statistical expectation, according to actual knowledge of variable y; and the belief state at time t is defined as in POMDP, $b_t = P(x_t \mid \Theta, \bar{c}_{t-1}, \bar{z}_t)$. The computational advantage of Eq. (8)

is that $Q_\Theta(c, b)$ can be directly obtained from the results of a POMDP solver, i.e. SARSOP (Kurniawati et al. 2008). We also provide a publicly available software and detailed implementation of our method, *http://doi.org/10.5281/zenodo.1161521*

Appendix C. Model estimation: computing the posterior of the model given new observations

At each time, after the agent takes an action, she receives an observation of the system. Although these observations are only noisy measurements of the state, they carry information about the models governing the dynamics of the system as well as accuracy of the measurements. In general, one might not be able to represent the posterior distribution of the model parameters given the historical observations and implemented policies in closed-form and as a result, need to adapt an approximate learning scheme based on Markov chain Monte Carlo (MCMC) (MacKay 2003). Although in this article, we assume that the uncertainty is among a set of candidate models (as in Walters and Hilborn (1976)) and as a result, learning can be done in closed-form using the Bayes rule. We update the knowledge of the model based on receiving a new piece of information at each time. Let $\Pi(\Theta)$ represent the prior knowledge about transition and observation models, $\Theta = \{T, O\}$. Suppose the agent takes action c_t at time t with the belief state $b_t(x)$, and receives an observation z_{t+1} in the next time step. The likelihood of this observation can be computed as follow:

$$P(z_{t+1} \mid b_t, c_t, \Theta) = \sum_{x' \in X} O(x', c_t, z_{t+1}) \sum_{x \in X} T(x, c_t, x') b_t(x) \tag{9}$$

Hence according to the Bayes Law, the posterior probability of the candidate model $\Pi(\Theta \mid z_{t+1})$ can be computed as,

$$\Pi(\Theta \mid z_{t+1}) \propto P(z_{t+1} \mid b_t, c_t, \Theta)\Pi(\Theta) \tag{10}$$

Appendix D. Computing the expected error in modeling the system dynamics

In order to evaluate the effectiveness of adaptive learning the model parameters, we introduce a metric called the expected error in approximating the system's dynamics, which is defined as follow,

$$e(t, \Pi(\Theta)) = D_{KL}(\Pi(\Theta_t) || \delta_{\Theta^*}) = \sum_{i=1}^{n} \ln \left(\frac{\Pi(\theta_{(t,i)})}{\delta_{\theta_i^*}} \right) \Pi(\theta_{(t,i)}) \tag{11}$$

where, D_{KL} is the Kullback-Leibler divergence (Cover and Thomas 2006), which is a non-symmetric measure of the differences between two probability distributions. δ_{Θ^*} is the Dirac delta function at the location of Θ^* (i.e. the true dynamics model), and $\Pi(\Theta_t)$ is the posterior distribution of the system's dynamics on the set of n candidate models $\Theta_t = \{\theta_{t,1}, ..., \theta_{t,n}\}$ at time step t.

At each time, after the learning is done we can calculate the KL divergence between the posterior distribution on the model and the true model as a metric to show the effectiveness of learning.

Appendix E. Additional figures

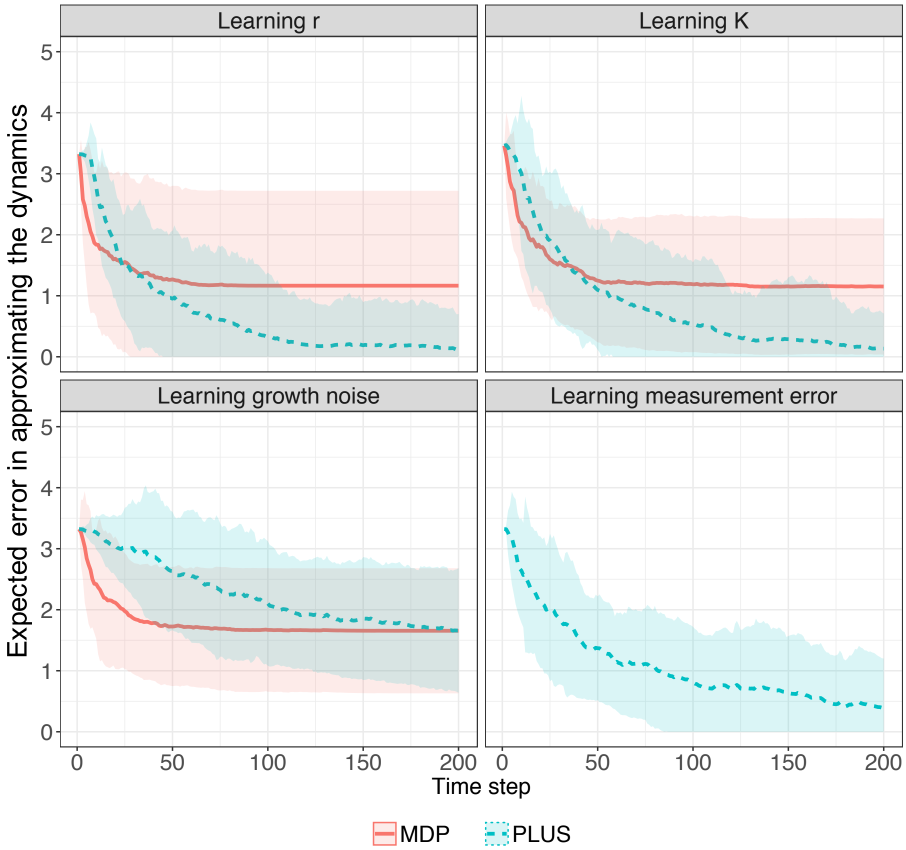

Figure S1: This figure shows the mean and +/- standard deviation of the expected error in approximating the true model describing the dynamics of the ecosystem by the two approaches of PLUS and Markov decision process (MDP). Results are based on 100 independent simulations and learning occurs after arrival of each new data. Candidate models for different parameters are 21 models for $r \in \{0.5, 1.5\}$, 11 models for $K \in \{0.5, 1\}$, and 10 models for $\sigma_t^X = \sigma_m \in \{0.05, 0.5\}$. Population dynamics is governed by the Beverton-Holt model, and the true model for each case is $r = 0.95$, $K = 0.75$, and $\sigma_t^X = \sigma_m = 0.25$. The measurement error in the process for learning r, K, and σ_t^X is assumed to be $\sigma_m = 0.5$

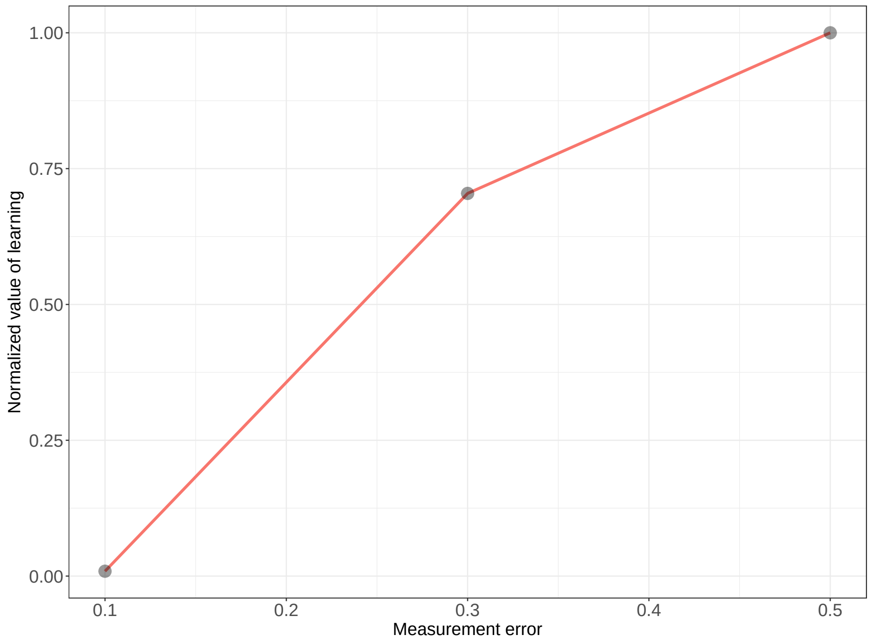

Figure S2: This figure shows the sensitivity of significance of learning to the measurement error. The model is governed by the Ricker model and the uncertainty is in the parameter r, the intrinsic productivity of the population. The main message of this figure is that as the measurement error increases, the value of learning the true model parameters is higher, which is intuitive. For example, when there is no uncertainty in measuring the state (measurement error is zero), the value of learning model parameters is lower, because full observability of the state includes maximum information for learning the model parameters.

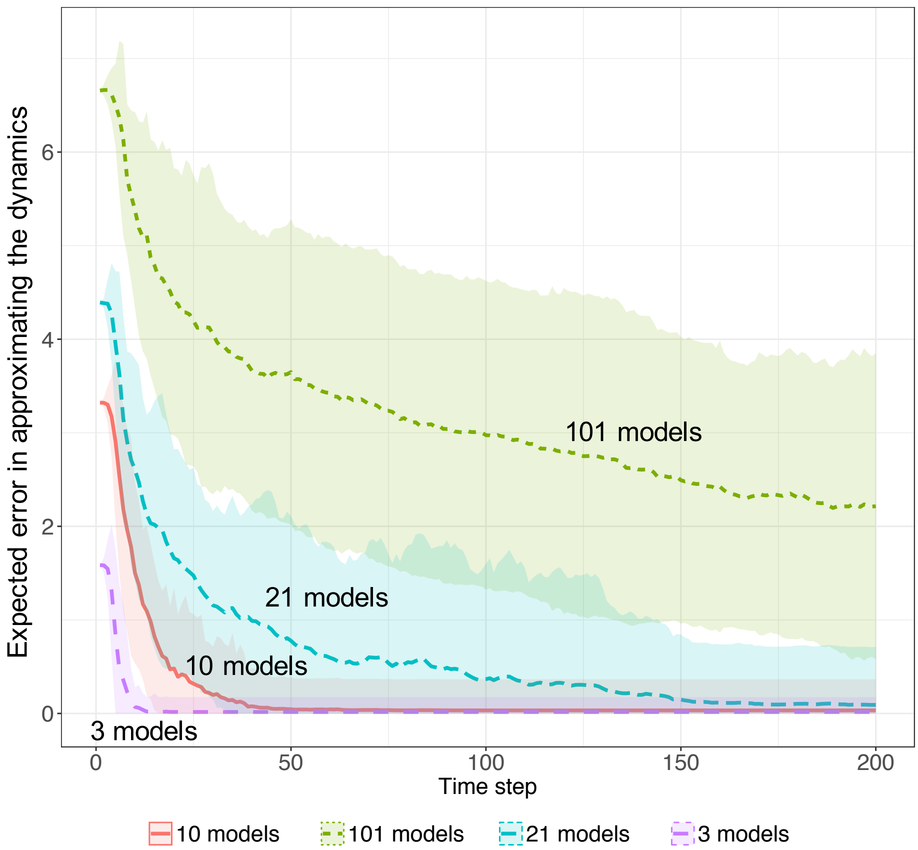

Figure S3: We evaluate the efficiency of learning process when there are 3, 10, 21, and 101 candidate models with $r \in \{0.5, 1.5\}$. Population dynamics of the models is modeled by the Ricker formula and the measurement error is assumed to be $\sigma_m = 0.5$. As we have less number of candidate models, it is easier to learn which model is the true model describing the dynamics of the system. For example, when we have 101 models, the difference between two candidate model's growth rate r is 0.01, which is very hard to distinguish and that is the reason that algorithm needs more data to exactly identify the true model and the learning process is slower. The figure shows mean and +/- standard deviations for 100 independent simulation.

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